SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

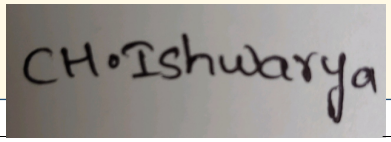
Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – INDIVIDUAL TALK

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 3 – INDIVIDUAL TALK
Weightage:	30 % of CIA	**Total Marks:**	20 marks
CO1 Statement:	*Analyse the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
Student Name:	Ch.Ishwarya	**Reg. No.:**	192324145
Section / Batch:	AI&DS	**Date:**	05/10/2026

Code of Conduct

I Ch.Ishwarya(192324145) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Studen 

Instructions

- Topic given before the session and students speak on stage.
- Duration: 10 minutes (In class)

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Individual Task: Access Channels with Mobile Communication

Introduction

Access channels are an important part of a mobile communication system. They provide the communication path between a mobile device and the base station. In simple words, access channels help a mobile phone request network access, establish communication, and exchange control information with the cellular network.

Basic Architecture

A mobile communication system can be represented as:

Mobile Phone → Access Channel → Base Station → Core Network → Other Networks

Access channels perform important functions such as:

- Requesting access to the network
- Establishing communication
- Sending control information
- Allocating communication resources
- Supporting mobility management
- Maintaining reliable communication

Main Types of Access Channels

1. Random Access Channel (RACH)
RACH is used by a mobile device to request access to the cellular network. It is commonly used when the mobile wants to initiate communication or respond to a paging request.

2. Paging Channel (PCH)
The Paging Channel is used by the network to notify a mobile device about an incoming call, message, or service request. The mobile listens to the paging information to determine whether it needs to respond.

3. Broadcast Control Channel (BCCH)
BCCH broadcasts important system information from the base station to mobile devices. It provides information such as network configuration, cell identity, and available communication parameters.

4. Access Grant Channel (AGCH)
AGCH provides a mobile device with information about the resources or channel it can use after making an access request**.**

5. Standalone Dedicated Control Channel (SDCCH)
SDCCH is used for signaling and control information before a dedicated traffic channel is assigned. It can support activities such as authentication and call setup.

Access Channels in 4G LTE

In 4G LTE, access procedures use channels such as:

- PRACH: Used by the mobile device to request initial access.
- PDCCH: Carries control information and resource allocation.
- PDSCH: Carries downlink user data and signaling.
- PUCCH: Carries uplink control information.
- PUSCH: Carries uplink user data.

These channels work together to provide efficient communication between the mobile device and the LTE base station.

Access Channels in 5G

In 5G, access procedures are designed to support high-speed communication, low latency, and a large number of connected devices. Important channels include:

- PRACH: Used for initial access and random access procedures.
- PDCCH: Carries downlink control information.
- PDSCH: Carries downlink user data.
- PUCCH: Carries uplink control information.
- PUSCH: Carries uplink user data.

5G access channels help devices establish connections quickly and efficiently with the network.

Conclusion

To conclude, access channels are essential for connecting mobile devices to cellular networks. They allow mobile users to request network access, receive system information, exchange control signals, and establish communication. From traditional 2G/3G channels to 4G LTE and 5G channels, access mechanisms have evolved to provide higher efficiency, faster access, lower latency, and support for a large number of mobile devices.

Geotag Photo:

